The arc is fine above 4 kHz. Below 2 kHz the room bends it by up to 8 dB.

For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height, in the propeller plane, around the propeller. A perfect setup reads the same level at every position: a circle. We did this twelve times (keys prop7–18, 1–2 Sept 2026, PWM 1900). What is left after we remove each microphone's own offset is the effect of the room.

2026-09-08
data: laptop backup of 2 Sept
script: scripts/arc_error_map.py
full guide: docs/arc-validation-guide.html

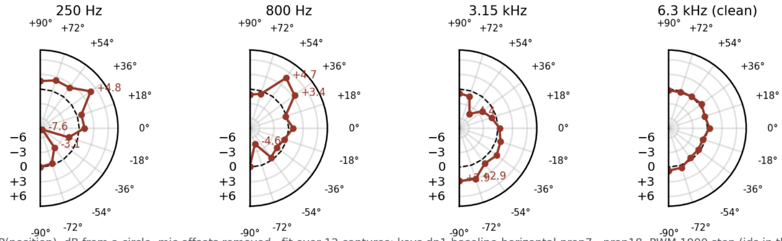


Fig. 1 · room error P(position), dB from a circle, mic offsets removed · fit over 12 captures: keys dp1-baseline-horizontal-prop7…prop18, PWM 1900 step (ids in the table below)

What the room does

The level at −72° is 2.4 to 5.7 dB too low between 500 and 2000 Hz, in all twelve runs. The dip is deepest at 630 and 800 Hz (−4.5 dB). It stays when a different microphone sits there (811-1896 in eight runs, 810-8901 in the four runs with the arc turned around). This position is the one closest to the chamber corner.

The level at +36° and +54° is 3 to 7 dB too high, at 200–250 Hz and at 630–1000 Hz. Three different microphones sat at +54° over the twelve runs. The bump stayed the same.

Around the blade tone (200–250 Hz) the two halves of the arc disagree: −36° reads −7.6 dB, +36° reads +4.8 dB; −54° reads −3 dB, +54° and +72° read +2 dB. The negative-angle half of the arc (−36°, −54°) is too low, the positive-angle half (+36°, +54°, +72°) is too high. The dip also moves with motor speed (−9 dB at PWM 1900, −2 dB at 2000). A sound wave that bounces off a surface and adds to the direct wave does exactly this: it cancels at some frequencies and adds at others, and the tone slides through those frequencies as the speed changes.

Above 4 kHz the arc reads a circle within ±1 dB at every position. The positions 0°, ±18° and ±90° are within about 1 dB in every band.

What it is not

Not the microphones, with one exception. Because the microphones at +72° and +54° were swapped after prop10, and the arc was turned around for prop11–14, we can tell apart what belongs to a microphone and what belongs to a position. Ten microphones agree within ±1.8 dB in every band. **Microphone 811-1892 reads 2–6 dB too low from 125 Hz to 2.5 kHz.** Its calibration file does not match the capsule (its sensitivity value, −10.63 dB, is the least negative of the set, while the raw capsule is only about 1 dB low). When it sat at +54° it made the room error look like 7 dB. Take it off the arc or measure it again.

Not a labelling error. prop11, 12, 13 and 14 were measured with the arc turned around. Their patterns match the mirror image of the other runs (correlation +0.79 to +0.98). Plotted by label, the dip appears on the other side. Plotted by physical position, they agree with the other eight runs.

Not one frequency. The error sits in three separate bands, 200–250 Hz, 630–1000 Hz and 3150 Hz, and nowhere else (Fig. 2). Rajmane and Baumann (DAGA 2016) found a steel plate in a chamber only shows up once the wavelength is shorter than the plate (35 cm from 1.25 kHz, 10 cm from 3.15 kHz). So three bands point to three object sizes: about 1.5 m (floor, table or wall), about 0.4 m (stand base or a bar of the arc frame), about 0.1 m (a microphone clamp or mount).

Fig. 2 · Microphone error map (left) and room error map (right), same fit, same scale

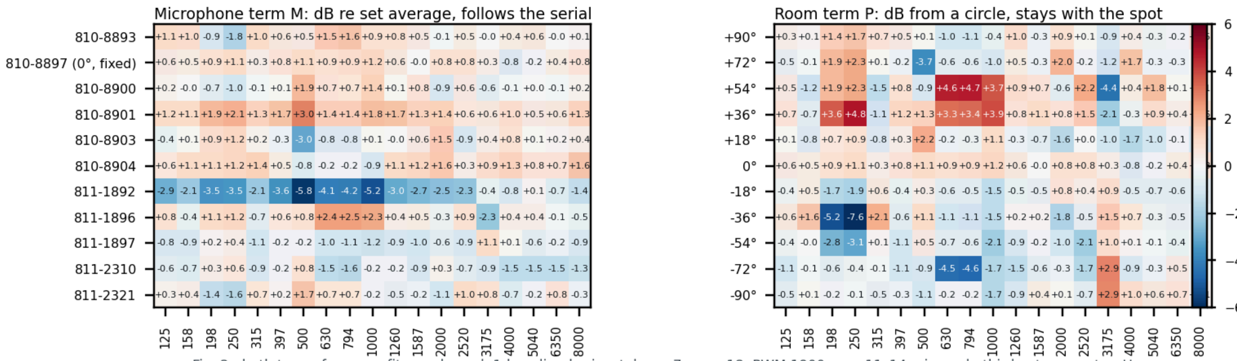
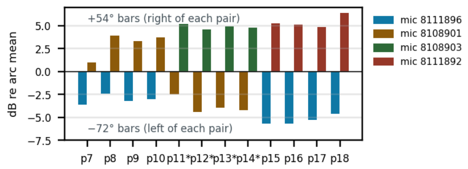


Fig. 2 · both terms from one fit over keys dp1-baseline-horizontal-prop7…prop18, PWM 1900, prop11–14 mirrored · third-octave centre, Hz

Fig. 3 · Two positions, twelve runs, three microphones



y prop7…18 (p7 = dp1-baseline-horizontal-prop7), PWM 1900 capture, * = arc turned 180°

500–2000 Hz, microphone offsets removed; * = arc turned around. The dip at −72° and the bump at +54° keep their size no matter which microphone sits there.

Where it points

With the arc flat, all microphones are at the same height. Floor and ceiling reflect the same into every microphone and cannot make one half of the arc differ from the other. The pattern we see is one-sided: the −36°/−54°/−72° side reads low, the +36°/+54° side reads high. So the surface is beside the propeller, in the horizontal plane: the wall, the corner or an object on the −angle side of the rig. The dip at 240 Hz that moves with motor speed means the bounced path is about 0.7 m longer than the direct one (half a wavelength), so that surface is close, within about half a metre of the microphones. The −72° dip at 630–1000 Hz (wavelength about 0.4 m) is a smaller object at the −90° end of the arc, on the corner side.

Next test (one afternoon)

1. Take 811-1892 off the arc.
2. Put a small loudspeaker on the stand instead of the propeller. Play short tone bursts at 250 and 800 Hz and record all microphones. The delay of the second arrival at each microphone, times 343 m/s, is the extra path length. Only one surface fits all eleven delays. With no airflow, anything left is the room.
3. Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the −90° end, distances from the −72° and +72° microphones to the nearest wall.
4. Cover or move that surface, then run scripts/arc_error_map.py again. Foam only helps at 3 kHz. At 250 Hz the only fix is distance.

Measurements plotted (all figures): the last PWM-1900 capture in each key `2004__6in__unset__dp1-baseline-horizontal-<key>`, 11 microphones × 2.0 s, calibrated; microphone serials 810-/811-xxxx given by their last four digits

KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop7	2026-09-01 16:42:31	norm	1892/8901
prop8	2026-09-01 16:44:39	norm	1892/8901
prop9	2026-09-01 16:50:45	norm	1892/8901
prop10	2026-09-01 16:55:35	norm	1892/8901
prop11	2026-09-02 13:18:49	180°	8901/1892
prop12	2026-09-02 14:24:12	180°	8901/1892
prop13	2026-09-02 14:34:37	180°	8901/1892
prop14	2026-09-02 14:37:02	180°	8901/1892
prop15	2026-09-02 15:48:59	norm	8901/1892
prop16	2026-09-02 15:54:26	norm	8901/1892
prop17	2026-09-02 16:52:44	norm	8901/1892
prop18	2026-09-02 17:08:19	norm	8901/1892

Left out: the 31 Aug keys and `prop1–prop6`; their patterns match neither arc orientation, so the rig or room was different then. "180°" = label +e was physically at −e.

How the numbers were made (not a sum, not an average)

Every level in the twelve runs (11 microphones × 19 third-octave bands × 12 runs) is written as three parts added together: **run gain + position error + microphone offset**. The three parts are found by least squares, i.e. the single set of values that fits all 12 × 11 levels best in every band. **Run gain** takes up any level difference between runs (RPM, day, load). **Microphone offset** takes up each serial's own error after calibration. **Position error** is what is left that belongs to a spot on the arc; it is forced to average zero over the eleven spots, so it reads as dB from a circle. That is what Figs. 1 and 2 show.

The turned-around runs are handled before the fit: for prop11–14 the label +e is entered as physical −e, so "−72°" always means the same spot in the room. The four microphone offsets and the eleven position errors can be told apart only because the same spot was covered by different microphones (the swap after prop10 and the four turned runs). Without that, a low microphone and a low spot would be indistinguishable. Fig. 3 is not from the fit: each bar is one run, level at the spot minus the mean of all eleven microphones in that run, with that microphone's fitted offset subtracted. Spectra: Welch 4096-point Hann, UMIK-2 calibration as the app applies it, third-octave sums 125 Hz–8 kHz. Code: `scripts/arc_error_map.py`, function `fit`; inputs and outputs in `docs/analysis/`.

Do the numbers add up?

Refitting from the data pack reproduces the published maps to 0.01 dB; each map sums to zero per band by construction. 132 measured cells per band are explained by 34 unknowns. 87% of all cells are within 2 dB of gain + P + M. Scatter is 0.6–0.8 dB in the quiet bands and 1.5–2.3 dB in 200–250 Hz and 500–1000 Hz, the bands where the room acts. The worst cells (+11 dB at −36°/250 Hz in prop7; +6 to +7 dB at 0°/2–4 kHz in prop7–8) are single tone-band cells in the first two runs where the blade tone sits on a band edge; they do not follow a microphone or a spot and move the maps by under 1 dB. Limit: the 0° microphone (810-8897) never moved, so its M and the 0° P are known only as a sum.

Worked example · −72° spot at 800 Hz, two mics, two orientations

RUN	MIC THERE	MEASURED	GAIN	P	M	MODEL	LEFT
prop18	811-1896	59.1	62.3	−4.6	+2.5	60.2	−1.1
prop12 (180°)	810-8901	58.6	62.2	−4.6	+1.4	59.0	−0.4

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4; prop18 with 811-1892 reads +3.1 = +4.7 − 4.2. Two mics, one room value.